Mode-Aware Anti-Jamming for Dynamic Spectrum Sharing UAV Networks:Joint Spectrum Allocation, Trajectory, and Power Optimization

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***Abstract*—Unmanned aerial vehicle (UAV) networks operating under dynamic spectrum sharing (DSS) are vulnerable to intelligent jamming, especially when adversaries adopt pattern-switching strategies that induce non-stationary interference. Such dynamics significantly degrade communication reliability and hinder fast recovery after interference transitions. To address this issue, a mode-aware anti-jamming framework for DSS-enabled UAV networks is proposed. A CNN–GRU based module is first designed to recognize jamming modes from short-term interference observations. The inferred mode information is then incorporated into a semantic-augmented Markov decision process. Based on this formulation, a hybrid reinforcement learning framework combining proximal policy optimization (PPO) and deep deterministic policy gradient (DDPG) is developed to jointly optimize spectrum allocation, UAV trajectory, and transmit power. Simulation results demonstrate that the proposed approach achieves faster adaptation to jamming mode transitions and significantly improves throughput and communication reliability compared with baseline schemes.**

***Index Terms*—Dynamic spectrum sharing, deep reinforcement learning, mode-aware anti-jamming, resource allocation**

I. Introduction

Recently, unmanned aerial vehicles (UAVs) have emerged as a promising solution for providing flexible and on-demand wireless connectivity in low-altitude networks. Benefiting from their high mobility and favorable line-of-sight (LoS) communication links, UAV-assisted systems have been widely applied in emergency communications, temporary hotspot coverage, and aerial relaying [1]–[3]. However, the limited availability of dedicated spectrum resources poses a fundamental challenge to UAV communications. To alleviate spectrum scarcity, dynamic spectrum sharing (DSS) has emerged as a promising paradigm, enabling UAV networks to opportunistically access licensed or unlicensed spectrum while coexisting with incumbent users [4]. Nevertheless, the open and dynamic nature of DSS environments makes UAV communications highly susceptible to interference, particularly in the presence of dense wireless devices and potential adversarial jamming. As a result, the wireless environment becomes highly dynamic and uncertain, posing significant challenges for efficient and adaptive resource allocation, including spectrum allocation, UAV trajectory optimization, and transmit power control.

The existing resource allocation approaches can be broadly categorized into two classes, namely, conventional optimization-based methods and intelligent learning-based methods, particularly those based on deep reinforcement learning (DRL). In [5], an alternating iterative optimization framework was developed to jointly optimize transmit power and UAV trajectory, where successive convex optimization was applied to trajectory design. In [6], the problem was extended to spectrum sharing scenarios, where joint trajectory and transmit power optimization was addressed using block coordinate descent (BCD) combined with successive convex optimization. Furthermore, more complex interference environments were considered in [7], where trajectory and power control under malicious jamming was solved via an alternating optimization (AO) framework with successive convex approximation (SCA) and semidefinite relaxation (SDR). Moreover, in [8], the problem was further generalized to mixed-integer formulations involving joint resource allocation and UAV trajectory control, which was tackled by decomposing it into subproblems and solving via alternating optimization. Despite their effectiveness, conventional optimization-based methods often lead to nonlinear and non-convex problems due to coupled variables and sensitivity to initial conditions, making them difficult to solve efficiently and prone to suboptimal solutions.

To overcome the limitations of conventional optimization-based methods, DRL-based approaches have gained increasing attention. In [9], a clustering-aided reinforcement learning approach was proposed for joint user association, UAV trajectory design, and power control. In [10], a DRL-based trajectory design scheme was developed for UAV communication systems against malicious jamming. Moreover, in [11], an online-offline MA2C-DDPG framework was proposed for dynamic spectrum sharing UAV networks, where spectrum allocation and UAV trajectory were jointly optimized. This direction was further advanced in [12], where an intelligent learning framework from external interaction to internal inference was developed for spectrum sharing and UAV trajectory optimization. In addition, in [13], a multi-agent DRL-based scheme was proposed for joint trajectory control, resource allocation, and user association in multi-fixed-wing UAV networks.

However, most existing DRL-based methods characterize interference only through observable states, such as channel conditions or received interference power, without explicitly modeling the underlying jammer behavior. For example, in [14], channel prediction was incorporated to improve resource allocation in dynamic spectrum-sharing networks,

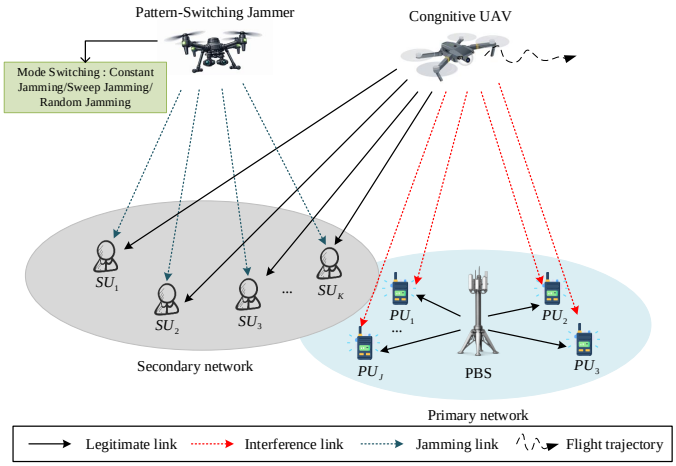

Fig. 1: The DSS-enabled UAV network under pattern-switching jamming.

while in [15], anti-jamming strategies were designed based on power and mobility control. Nevertheless, pattern-switching jamming, where interference modes change abruptly over time, has rarely been explicitly considered, leading to slow adaptation in highly non-stationary environments.

In this paper, a mode-aware anti-jamming framework is developed for DSS-enabled UAV networks under pattern-switching interference. Unlike conventional approaches that rely on implicit interference learning, A CNN–GRU based module is first introduced to recognize jamming modes from short-term interference observations and capture the dynamic interference patterns. By integrating the inferred jamming mode into the Markov decision process, the proposed framework enables faster adaptation to non-stationary environments. Furthermore, a hybrid reinforcement learning framework combining PPO and DDPG is designed to address the coupled discrete–continuous optimization problem, where spectrum allocation, UAV trajectory, and transmit power are jointly optimized. Extensive simulation results verify that the proposed framework significantly improves communication reliability and spectrum efficiency, while achieving faster recovery after jamming mode transitions compared with baseline schemes.

The remainder of this paper is organized as follows. The system model is presented in Section II. Section III presents our proposed mode-aware anti-jamming framework. Section IV presents the simulation results. Finally, the paper concludes with Section V.

II. System Model

A. Scenario Description

We consider a UAV-enabled dynamic spectrum sharing (DSS) network under intentional pattern-switching jamming, as illustrated in Fig. 1. The network consists of a primary network with one primary base station (PBS) and J primary users (PUs), and a cognitive UAV-assisted secondary network serving K ground secondary users (SUs) over M licensed sub-bands. Each PU occupies one orthogonal licensed sub-band, while the C-UAV opportunistically accesses the spectrum to provide downlink transmission for the SUs under PU protection constraints. Meanwhile, a jamming UAV (J-UAV) interferes with the secondary transmission by switching among different jamming patterns over time.

A three-dimensional Cartesian coordinate system is considered, where the UAVs fly at a fixed altitude H_c. The horizontal locations of the PBS, the j-th PU, the k-th SU, and the C-UAV at time slot n are denoted by $\mathbf{w}_b$, $\mathbf{w}_{p,j}$, $\mathbf{w}_{s,k}$, and $\mathbf{q}_c[n]$, respectively. The total communication duration T is divided into N equal-length time slots with duration $\Delta t = T/N$. The ground SUs move according to slot-wise kinematic updates, while the horizontal position of the C-UAV evolves as

$$\begin{aligned} x_c[n+1] &= x_c[n] + v_c \cos\big(\phi_c[n]\big), \\ y_c[n+1] &= y_c[n] + v_c \sin\big(\phi_c[n]\big), \end{aligned} \tag{1}$$

where v_c and $\phi_c[n]$ denote the flight speed and heading angle of the C-UAV, respectively. Accordingly, the distance between the C-UAV and the k-th SU at time slot n is

$$d_{c,k}[n] = \sqrt{\big\|\mathbf{q}_c[n] - \mathbf{w}_{s,k}[n]\big\|^2 + H_c^2}. \tag{2}$$

The air-to-ground channel between the C-UAV and the SUs is dominated by line-of-sight (LoS) propagation. Let β_{ref} denote the channel power gain at the reference distance of 1 m. Then, the channel power gain from the C-UAV to the k-th SU is given by

$$h_{C,k,n}^{\text{LoS}} = \beta_{\text{ref}} d_{c,k}[n]^{-2}. \tag{3}$$

For the terrestrial links, distance-dependent path loss and Rayleigh fading are considered.

Unlike conventional fixed or random jamming models, the J-UAV adopts a pattern-switching strategy. Let $z[n] \in \mathcal{Z} \triangleq \{\text{Constant}, \text{Sweep}, \text{Random}\}$ denote the jamming mode at time slot n, corresponding to constant jamming, sweep jamming, and random jamming, respectively. The jamming mode evolves as a first-order Markov chain, i.e.,

$$\Pr(z[n+1] = z' \mid z[n] = z) = \Pi_{z,z'}, \quad z, z' \in \mathcal{Z}, \tag{4}$$

where $\mathbf{\Pi}$ is the mode transition probability matrix. Since the transition probabilities are unknown and the instantaneous jamming mode cannot be directly observed by the C-UAV, the resulting interference is time-varying and non-stationary.

To characterize spectrum occupancy, let $\rho_{k,n}[m] \in \{0,1\}$ and $\rho_{J,n}[m] \in \{0,1\}$ denote the sub-band usage indicators of the k-th SU and the J-UAV, respectively. Specifically, $\rho_{k,n}[m] = 1$ indicates that sub-band m is assigned to the k-th SU at time slot n, while $\rho_{J,n}[m] = 1$ means that the same sub-band is jammed at slot n. Based on the sub-band availability and the time-varying jamming condition, the C-UAV dynamically adjusts spectrum allocation to restore communication quality after jamming pattern changes while protecting the primary network.

Accordingly, the SINR of the k-th SU on sub-band m at time slot n is expressed as

$$\gamma^{s}_{m,k,n} = \frac{\rho_{k,n}[m] P_C[n] h^{\mathrm{LoS}}_{C,k,n}}{\sigma^2 + P_B h^{\mathrm{NLoS}}_{P,k,n} + \rho_{J,n}[m] P_J h^{\mathrm{LoS}}_{J,k,n}}, \tag{5}$$

where P_B and P_J denote the transmit powers of the PBS and the J-UAV, respectively, $P_C[n] \in [P_C^{\min}, P_C^{\max}]$ is the transmit power of the C-UAV at time slot n, and σ^2 is the noise power. The corresponding achievable rate is

$$R^{s}_{m,k,n} = B \log_2\big(1 + \gamma^{s}_{m,k,n}\big), \tag{6}$$

where B is the bandwidth of each sub-band. The achievable PU rate is defined similarly and is required to satisfy a minimum-rate constraint.

To capture communication reliability under pattern-switching jamming, an outage event is declared when the achievable SU rate falls below a QoS threshold R_{th}. Accordingly, the outage indicator is defined as

$$\mathbb{I}^{\mathrm{out}}_{k,n} \triangleq \mathbf{1}\big\{R^{s}_{k,n} < R_{\mathrm{th}}\big\}, \quad \forall k \in \mathcal{K},\ \forall n \in \mathcal{N}, \tag{7}$$

where $\mathbf{1}\{\cdot\}$ equals 1 if the condition holds and 0 otherwise. This definition penalizes each outage slot and thus encourages fast recovery after jamming mode transitions.

B. Problem Formulation

In this paper, we address a joint spectrum allocation, UAV trajectory control, and transmit power adaptation problem in a dynamic spectrum sharing UAV network under pattern-switching jamming, aiming to achieve fast recovery and reliable communication for secondary users while satisfying the primary user protection constraints. Accordingly, the optimization problem is formulated as follows

$$\textbf{P1:} \quad \max_{\boldsymbol{\rho},\,\boldsymbol{\phi},\,\mathbf{P}} \quad \frac{1}{N}\sum_{n=1}^{N}\sum_{k=1}^{K}\sum_{m=1}^{M} R^{\mathrm{s}}_{m,k,n} - \frac{\lambda_{\mathrm{out}}}{N}\sum_{n=1}^{N}\sum_{k=1}^{K}\mathbb{I}^{\mathrm{out}}_{k,n} \tag{8a}$$

$$\text{s.t. } \textbf{C1:} \quad \sum_{m=1}^{M} R^{p}_{m,j,n} \geq R^{\min}_{j}, \quad \forall j \in \mathcal{J},\ \forall n \in \mathcal{N} \tag{8b}$$

$$\textbf{C2:} \quad \rho_{k,n}[m] \in \{0,1\}, \quad \forall k \in \mathcal{K},\ \forall m \in \mathcal{M},\ \forall n \in \mathcal{N} \tag{8c}$$

$$\textbf{C3:} \quad \sum_{k=1}^{K} \rho_{k,n}[m] \leq 1, \quad \forall m \in \mathcal{M},\ \forall n \in \mathcal{N} \tag{8d}$$

$$\textbf{C4:} \quad |\phi_c[n]| \leq \pi, \quad \forall n \in \mathcal{N} \tag{8e}$$

$$\textbf{C5:} \quad P_C^{\min} \leq P_C[n] \leq P_C^{\max}, \quad \forall n \in \mathcal{N} \tag{8f}$$

The objective jointly accounts for throughput and communication reliability by maximizing the achievable rate of the SUs while penalizing outage events. Here, λ_{out} controls the tradeoff between rate maximization and outage suppression, so that prolonged service interruption under dynamic jamming is discouraged. In (8), $\boldsymbol{\rho}$, $\boldsymbol{\phi}$, and $\mathbf{P}$ denote the sub-band allocation, UAV heading control, and C-UAV transmit power policies, respectively. Constraint (8b) guarantees PU protection, constraints (8c) and (8d) enforce feasible orthogonal spectrum allocation among SUs, and constraints (8e) and (8f) ensure feasible UAV maneuvering and transmit power adaptation.

III. MODE-AWARE HYBRID PPO-DDPG FRAMEWORK FOR ANTI-JAMMING UAV COMMUNICATIONS

A. Jamming Mode Recognition

The jammer switches among finite modes $z[n] \in \mathcal{Z}$ following the Markov model in (4). At each time slot, the C-UAV performs spectrum sensing and collects the interference-energy observation $\mathbf{I}[n] \in \mathbb{R}^M$ over the M sub-bands. Based on the most recent L slots, a length-L sensing window is constructed as

$$\mathcal{I}[n] = \{\mathbf{I}[n-L+1], \ldots, \mathbf{I}[n]\} \in \mathbb{R}^{L\times M}. \tag{9}$$

A lightweight recognizer $f_\theta(\cdot)$ maps $\mathcal{I}[n]$ to a mode probability vector

$$\boldsymbol{\pi}[n] = f_\theta(\mathcal{I}[n]) \in \mathbb{R}^{|\mathcal{Z}|}, \quad \sum_{c=1}^{|\mathcal{Z}|} \pi_c[n] = 1, \tag{10}$$

which provides a semantic description of the current jamming behavior. A CNN–GRU classifier is adopted, where a 1D CNN extracts spectral features and a GRU captures short-term temporal dynamics. The model is trained offline and used online only for inference.

B. Semantic-Augmented MDP Formulation

Following the MDP formulation in the reference work [11], we incorporate the mode probability vector $\boldsymbol{\pi}[n]$ as a semantic descriptor of the interference environment to facilitate fast adaptation under mode-switching jamming.

State: The system state at slot n is defined as

$$\mathbf{s}[n] = \big(\mathbf{a}[n-1],\, \mathbf{q}_c[n-1],\, \mathbf{g}[n],\, \boldsymbol{\pi}[n]\big), \tag{11}$$

where $\mathbf{a}[n-1]$ denotes the previous action at slot $n-1$, $\mathbf{q}_c[n-1]$ denotes the previous UAV position at slot $n-1$, $\mathbf{g}[n]$ denotes channel and geometry-related features for rate evaluation, and $\boldsymbol{\pi}[n]$ encodes the high-level semantic belief of the current jamming behavior.

Action: The action consists of a discrete sub-band allocation decision and a continuous control vector,

$$\mathbf{a}[n] = \big(\mathbf{a}_d[n],\, \mathbf{a}_c[n]\big), \tag{12}$$

where $\mathbf{a}_d[n]$ denotes the joint sub-band allocation for all K SUs at slot n, which is equivalently represented by the binary allocation variables $\{\rho_{k,n}[m]\}$ in the optimization problem. The continuous action is defined as

$$\mathbf{a}_c[n] = \big(\phi_c[n],\, P_C[n]\big), \tag{13}$$

where $\phi_c[n] \in [-\pi, \pi]$ and $P_C[n] \in [P_C^{\min},\, P_C^{\max}]$ denote the UAV heading angle and transmit power, respectively.

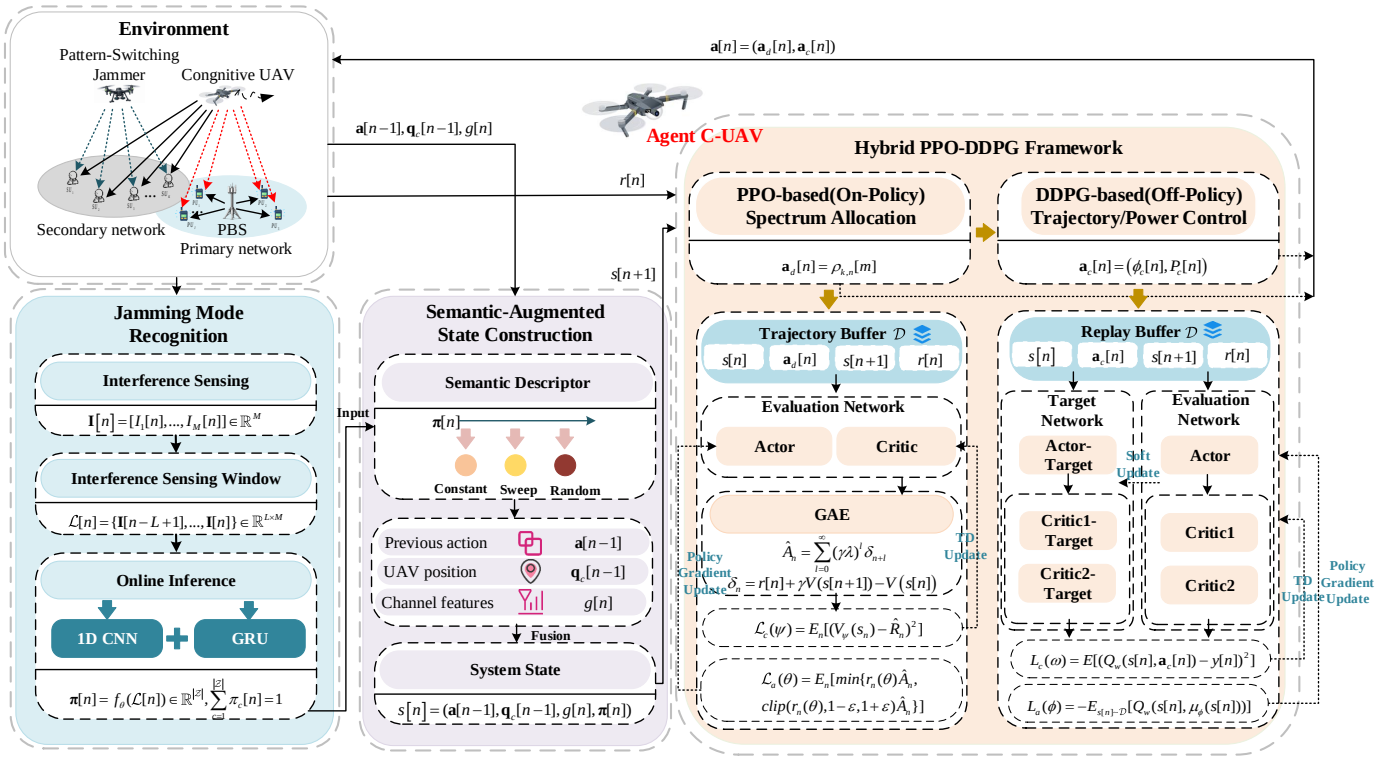

Fig. 2: The mode-aware hybrid PPO-DDPG framework for joint spectrum allocation, UAV trajectory, and power control.

Reward: Consistent with (8), we adopt a reliability-aware reward that jointly captures the secondary throughput, outage reliability, and PU protection, defined as

$$r[n] = \alpha \sum_{k=1}^{K} R_k^s[n] + \beta \sum_{j=1}^{J} \delta_j[n] - \gamma \sum_{k=1}^{K} \mathbb{I}(R_k^s[n] < R_{\text{th}}), \tag{14}$$

where $\alpha, \beta, \gamma > 0$ are constant coefficients, $\delta_j[n]$ is a penalty term when the interference to the PUs exceeds the tolerated threshold and formulated as

$$\delta_j[n] = \begin{cases} 0, & R_j^p[n] \geq R_j^{\min}, \\ R_j^p[n] - R_j^{\min}, & 0 \leq R_j^p[n] < R_j^{\min}, \end{cases} \tag{15}$$

where $R_j^p[n]$ is the achievable rate of the j-th PU and $R_j^{\min}$ is its minimum rate requirement.

C. Hybrid PPO-DDPG Framework

Problem (8) involves discrete spectrum allocation, continuous joint control of UAV mobility and transmit power. Thus, we employ a hybrid solver, as shown in Fig. 2, where PPO is adopted for the discrete spectrum decision $\mathbf{a}_d[n]$ and DDPG is used for the continuous control vector $\mathbf{a}_c[n]$, both conditioned on the semantic-augmented state $\mathbf{s}[n]$.

PPO for Spectrum Allocation: Let $\pi_\theta(\mathbf{a}_d|\mathbf{s})$ denote the stochastic policy for spectrum allocation parameterized by θ, where $\mathbf{a}_d[n]$ is the discrete joint sub-band allocation decision at time slot n. Different from the multi-agent A2C design in [11], we adopt a centralized PPO-based spectrum allocator. To avoid the exponential growth of the joint discrete action space, the policy is implemented in a factorized multi-discrete form as

$$\pi_\theta(\mathbf{a}_d|\mathbf{s}) = \prod_{k=1}^{K} \pi_\theta^{(k)}(a_k|\mathbf{s}), \tag{16}$$

where each $\pi_\theta^{(k)}(\cdot|\mathbf{s})$ is a categorical distribution over the M candidate sub-bands. To satisfy the orthogonal sub-band constraint, action masking is adopted during action selection to prevent infeasible allocations and ensure that each sub-band is assigned to at most one SU per slot. PPO then updates the discrete allocation policy by maximizing the expected return while preventing excessively large policy changes. Specifically, the clipped surrogate objective is given by

$$\mathcal{L}^{\text{CLIP}}(\theta) = \mathbb{E}_n\Big[\min\Big\{r_n(\theta)\hat{A}_n, \\ \text{clip}(r_n(\theta),\, 1-\epsilon,\, 1+\epsilon)\,\hat{A}_n\Big\}\Big], \tag{17}$$

where $r_n(\theta) = \frac{\pi_\theta(\mathbf{a}_d[n]|\mathbf{s}[n])}{\pi_{\theta_{\text{old}}}(\mathbf{a}_d[n]|\mathbf{s}[n])}$ is the probability ratio between the new and old policies, ϵ is the clipping parameter, and $\hat{A}_n$ denotes the advantage estimate. In our implementation, the advantage is estimated to improve training stability and sample efficiency.

DDPG for Trajectory/Power Control: Given the semantic-augmented state $\mathbf{s}[n]$, we employ DDPG to learn the continuous control policy in an off-policy manner. Specifically, the transition tuple $(\mathbf{s}[n], \mathbf{a}_c[n], r[n], \mathbf{s}[n+1])$ is stored in a replay buffer, and mini-batches are uniformly sampled to update the

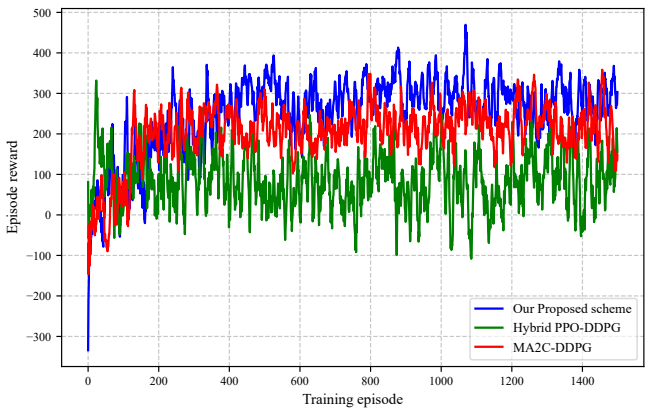

Fig. 3: Convergence comparisons of our proposed scheme with baseline schemes.

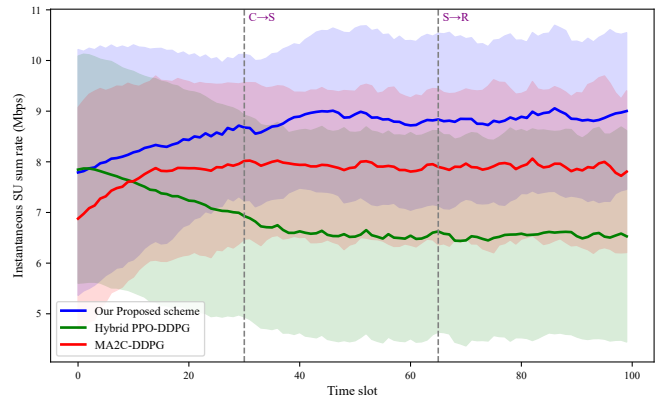

Fig. 4: Recovery comparisons of our proposed scheme with baseline schemes under jammer mode transitions.

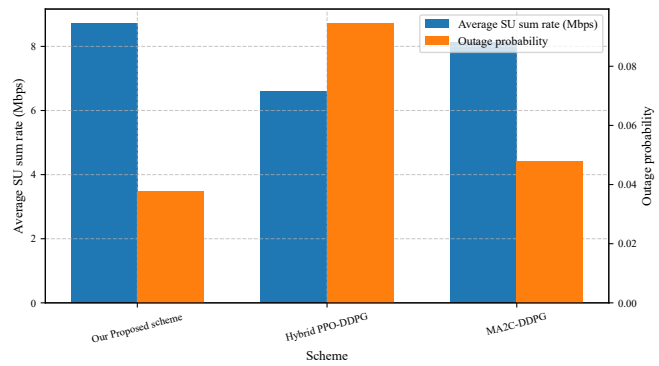

Fig. 5: Overall throughput and reliability comparisons of our proposed scheme with baseline schemes.

critic network $Q_\omega(\mathbf{s}, \mathbf{a}_c)$. For each sampled transition, the bootstrap target is constructed by the target networks as

$$y[n] = r[n] + \gamma\, Q_{\bar{\omega}}\Big(\mathbf{s}[n+1],\, \mu_{\bar{\varphi}}(\mathbf{s}[n+1])\Big), \tag{18}$$

and the critic is trained by minimizing the temporal-difference loss

$$L_c(\omega) = \mathbb{E}\Big[\big(Q_\omega(\mathbf{s}[n], \mathbf{a}_c[n]) - y[n]\big)^2\Big]. \tag{19}$$

The actor is then updated to maximize the critic-evaluated Q-value, while the corresponding target networks are updated via Polyak averaging. At each time slot, PPO outputs the discrete spectrum allocation $\mathbf{a}_d[n]$ (on-policy), and DDPG outputs the continuous control action $\mathbf{a}_c[n]$ (off-policy) based on $\mathbf{s}[n]$.

IV. SIMULATION RESULTS

In this section, simulation results are presented to evaluate the performance of the proposed joint spectrum allocation, trajectory optimization, and power control scheme. The simulation settings are based on the works in [11]. Specifically, SUs are PUs are randomly deployed in two seperate $500\,\mathrm{m} \times 500\,\mathrm{m}$ cells with a C-UAV flying at a fixed altitude of $100\,\mathrm{m}$. The number of SUs is 4. The transmit power of C-UAV ranges from 200mW to 300mW. The sensing window length for mode recognition is $L = 5$. The transition matrix is set as

$$\mathbf{\Pi} = \begin{bmatrix} 0.85 & 0.10 & 0.05 \\ 0.10 & 0.80 & 0.10 \\ 0.10 & 0.10 & 0.80 \end{bmatrix}. \tag{20}$$

Two baseline schemes are introduced for performance comparison. **MA2C-DDPG:** MA2C and DDPG are used for spectrum allocation and trajectory optimization, with the transmit power of the C-UAV fixed at 250 mW. **Hybrid PPO-DDPG:** Hybrid PPO-DDPG framework is adopted without the jamming mode recognition module.

As shown in Fig. 3, the convergence of the proposed scheme is presented with respect to training episodes. It is seen that all three schemes exhibit an overall increasing trend and gradually converge, while noticeable reward fluctuations appear in the first 150–200 episodes due to random exploration and jammer mode switching. After around 400 episodes, the training curves become relatively stable, indicating that effective anti-jamming policies have been learned. Moreover, the proposed scheme converges faster and reaches the highest reward among all compared methods, since the introduced mode-aware state representation reduces the ambiguity caused by non-stationary interference and makes policy learning more efficient. In contrast, Hybrid PPO-DDPG performs worse than MA2C-DDPG, showing that increasing algorithmic complexity alone is insufficient when the interference dynamics are not effectively captured.

Fig. 4 shows the average instantaneous SU sum rate under a representative jammer mode-switching sequence, where the solid curves denote the mean performance over multiple test episodes and the shaded regions represent one standard deviation. It is observed that the proposed scheme maintains the highest sum-rate level over most time slots. Moreover, around the jammer mode transition points, it exhibits smaller performance degradation and recovers more quickly than the baseline schemes. This is because the proposed framework

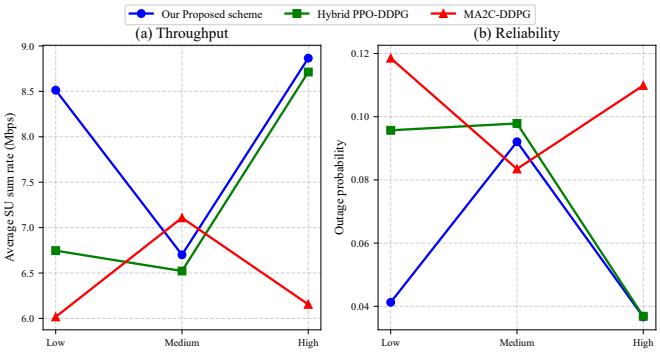

Fig. 6: Robustness comparisons of our proposed scheme with baseline schemes under different jammer transition settings

can infer the underlying jammer pattern from short-term observations, allowing the agent to adjust its behavior more promptly when the interference mode changes. Therefore, the proposed scheme achieves both better transient robustness and stronger adaptability under pattern-switching jamming.

Fig. 5 shows the overall performance comparison of the proposed scheme with the baseline schemes in terms of average SU sum rate and outage probability. It is seen that the proposed scheme achieves the highest average SU sum rate and the lowest outage probability among all compared methods. This indicates that the proposed mode-aware framework can make more effective long-term decisions under dynamic interference, thereby improving spectral efficiency while maintaining more reliable communication links. Therefore, compared with Hybrid PPO-DDPG and MA2C-DDPG, the proposed scheme achieves a better throughput–reliability tradeoff in DSS-enabled UAV networks.

The performance comparison under different jammer transition settings is shown in Fig. 6, where each scheme is trained and tested under the corresponding transition matrix. It is observed that the performance does not vary monotonically from Low to High, indicating that the impact of jammer transition statistics is determined not only by the switching frequency, but also by the resulting temporal structure and dwell-time distribution of different jamming modes. In particular, MA2C-DDPG exhibits a different trend from the proposed scheme and Hybrid PPO-DDPG, which suggests that different learning frameworks respond differently to the underlying jammer dynamics. Nevertheless, the proposed scheme maintains competitive throughput and reliability across all considered settings, demonstrating stronger adaptability to different jammer transition environments.

V. Conclusion

This paper investigated mode-aware anti-jamming communication in dynamic spectrum sharing UAV networks under pattern-switching interference. A mode-aware anti-jamming framework was developed in which the UAV first recognizes the current jamming pattern using a CNN–GRU based classifier and then incorporates the inferred mode information into the decision-making process. By formulating a semantic-augmented MDP, a hybrid PPO–DDPG reinforcement learning algorithm was proposed to jointly optimize spectrum allocation, UAV trajectory, and transmit power. Simulation results demonstrated that the proposed approach can effectively adapt to dynamic interference environments, achieve faster recovery after jamming mode transitions, and significantly improve both average throughput and communication reliability compared with baseline schemes. The proposed framework provides a promising solution for robust UAV communications in adversarial spectrum-sharing environments.

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